

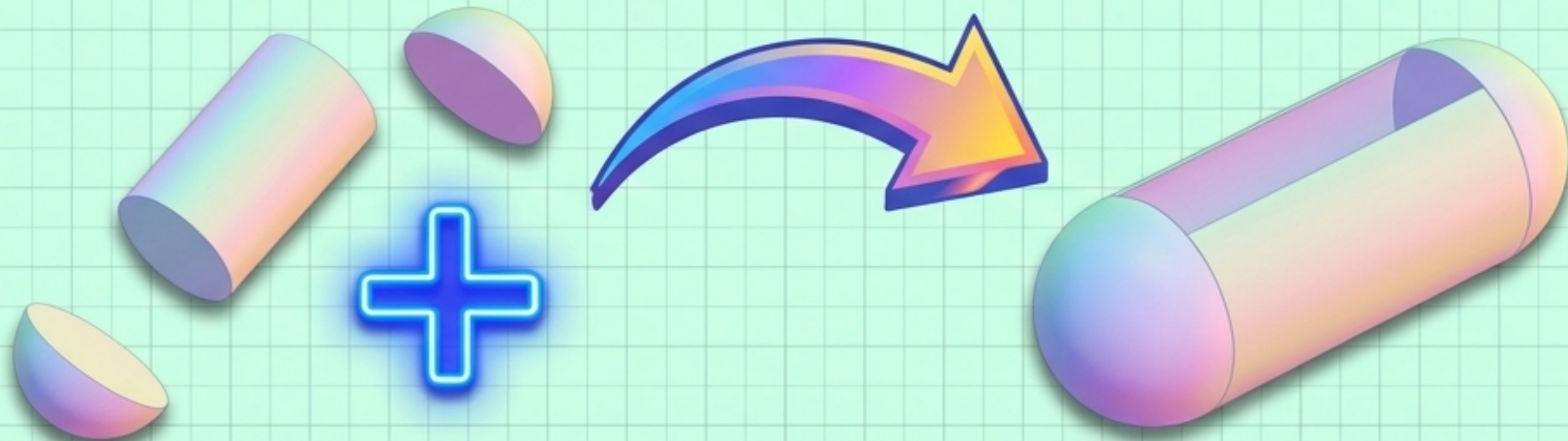
Mastering Surface Areas & Volumes

THE AESTHETIC GUIDE TO COMBINING SOLIDS! 🎀 ✨



The Golden Rule ✨

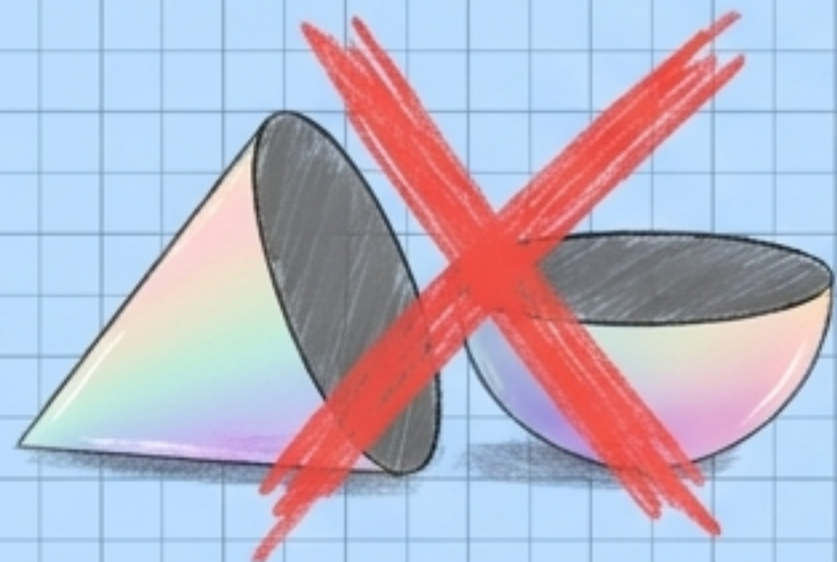
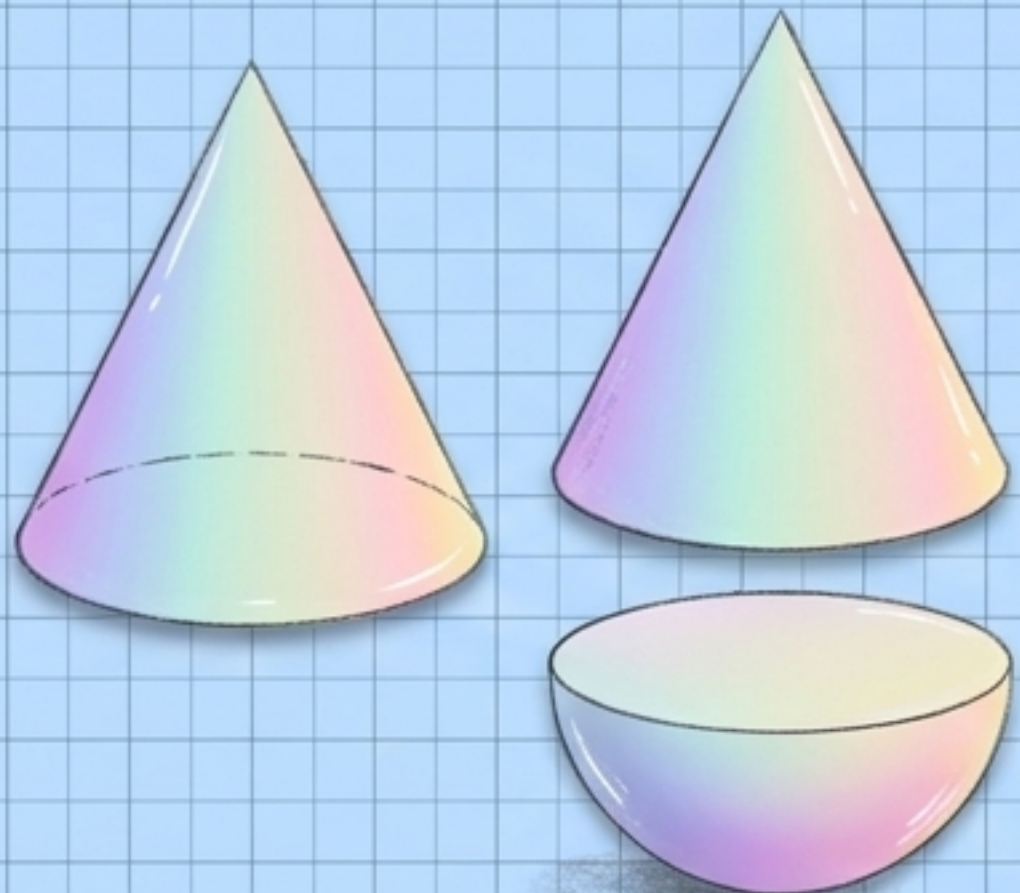
Don't panic when you see a weird shape! Every complex solid is just basic 3D shapes playing dress-up. Break them apart to find their secrets.



Surface Area: The Paintbrush Test 🎨

WARNING!

TSA of the new solid is **NOT** the sum of the TSAs of the parts!



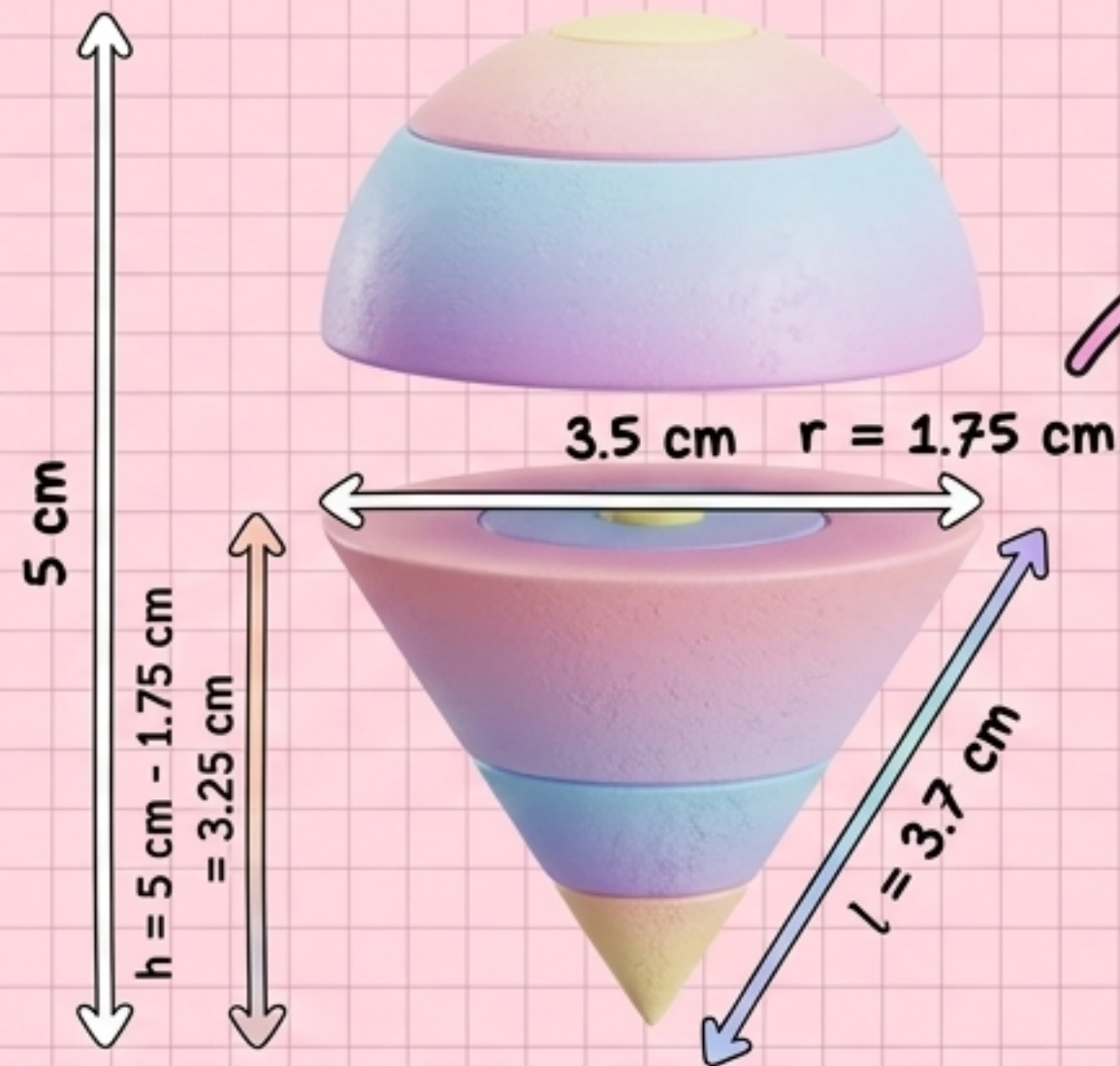
When shapes touch, their inner faces disappear! You can only paint the outside. We only care about Curved Surface Areas (CSA)!



$$\text{TSA}_{\text{total}} = \text{CSA}_1 + \text{CSA}_2$$

Let's Build a Lattu!

Total Height = 5cm,
Diameter = 3.5cm.

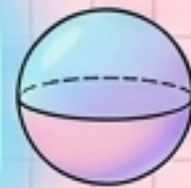


Find the slant height (l)
using Pythagoras!

$$l = \sqrt{r^2 + h^2} = 3.7 \text{ cm.}$$



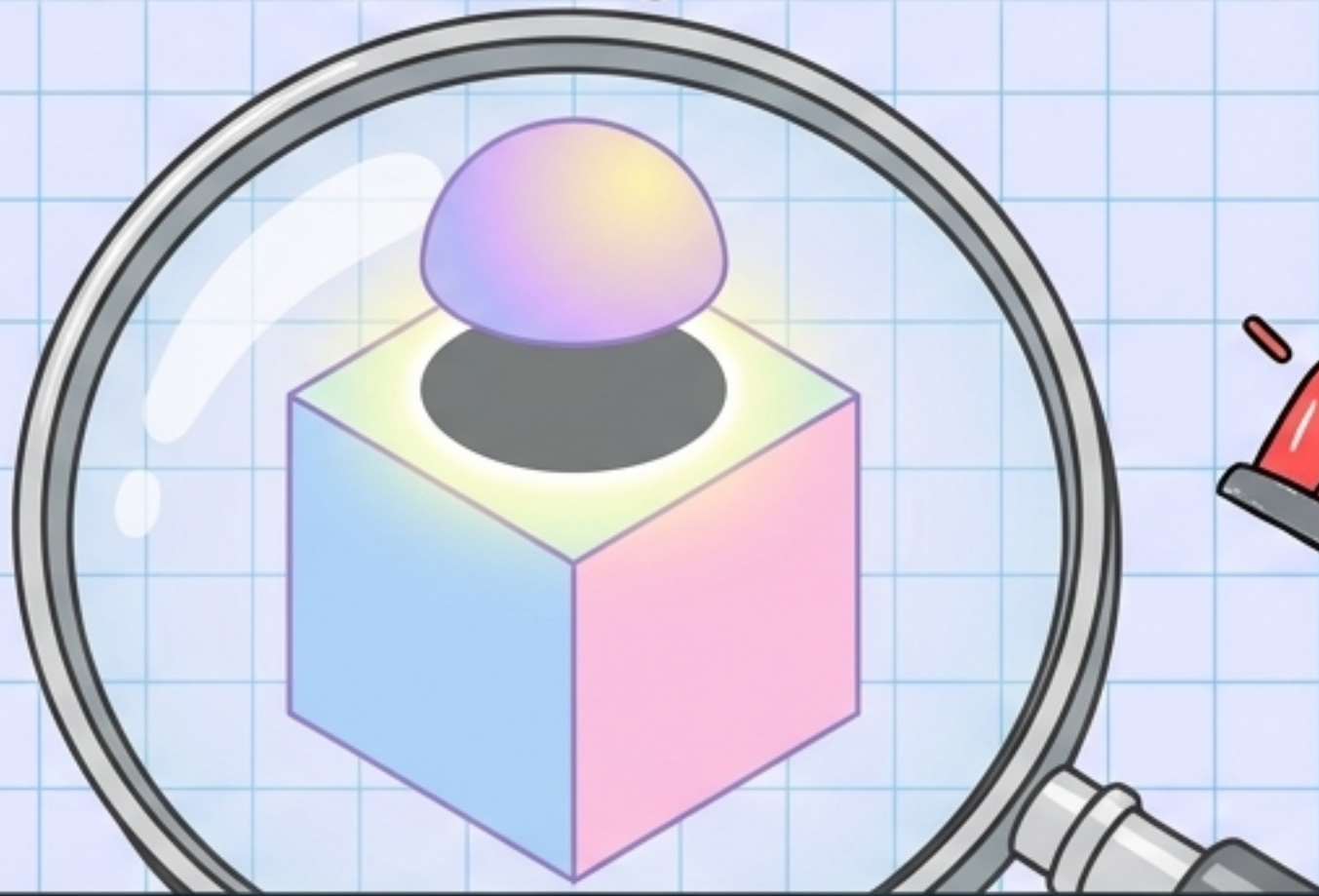
$$\text{TSA} = \pi r l \text{ (Cone)} + 2\pi r^2 \text{ (Hemisphere)}$$



Watch out for the Overlap Trap!



What happens when the base of Shape A is bigger than Shape B?
The smaller shape covers up a ring or circle of the bigger shape!

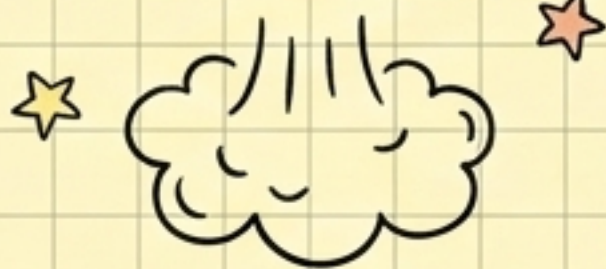


**Total Area = TSA of Cube - Base of Hemisphere
+ CSA of Hemisphere.**

Level Up: The Toy Rocket

Base of the cone (5cm) is larger than the cylinder (3cm). We have a leftover ring!

$$\text{Area} = \text{CSA}_{\text{cone}} + \text{Base}_{\text{cone}} - \text{Base}_{\text{cylinder}}$$



Always ask:
Can my paintbrush
touch this?

Only the bottom is
exposed!

$$\text{Area} = \text{CSA}_{\text{cylinder}} + \text{Bottom Base}$$



Volume: The Water Test



Volume is super straightforward. It's all about capacity—how much 3D space or water fits inside. Hidden faces don't matter at all!



Stacking shapes?
Just **ADD** their
volumes!
 $(V_{-1} + V_{-2})$

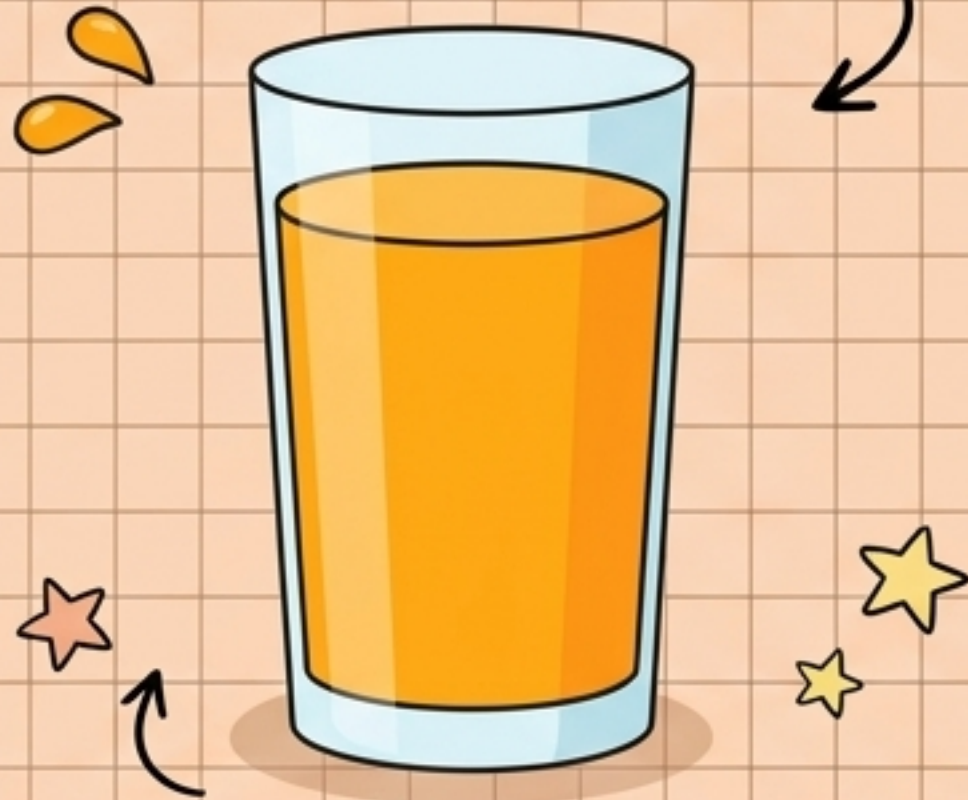


Scooping out a
shape?
Just **SUBTRACT**
its volume!
 $(V_{\text{total}} - V_{\text{scoop}})$



THE SNEAKY JUICE SELLER

EXPECTATION



APPARENT CAPACITY = $\pi r^2 h$
(FULL CYLINDER)

REALITY



ACTUAL CAPACITY =
CYLINDER VOLUME - HEMISPHERE VOLUME

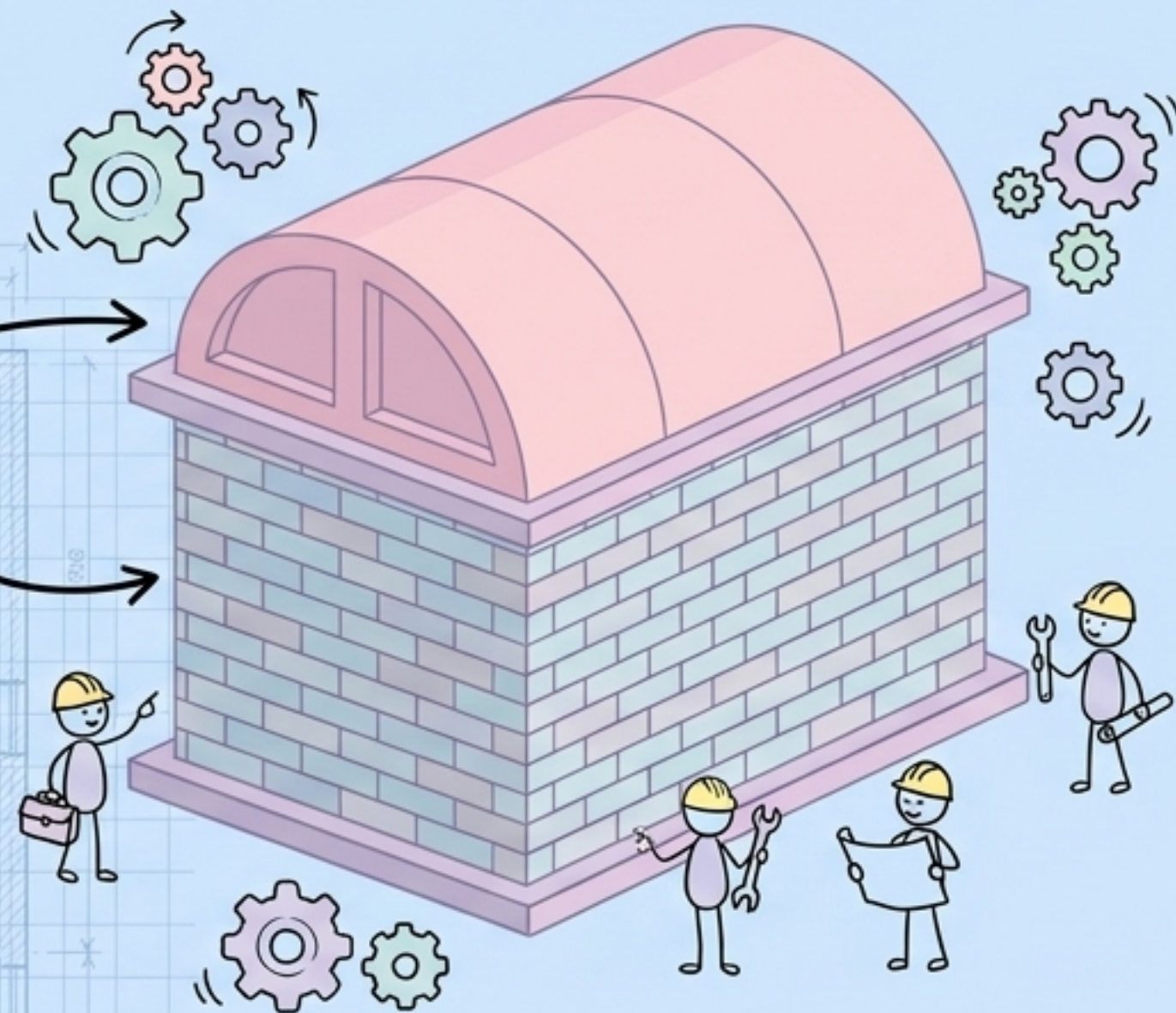
SCOOPIING OUT A HEMISPHERE FROM THE BOTTOM
REDUCES THE CAPACITY! WATER TEST = SUBTRACT!

Volume in the Real World

$$\text{Total Air Space} = \text{Volume of Cuboid} + \left(\frac{1}{2} \times \text{Volume of Cylinder} \right)$$

Wait! What about the stuff inside?

$$\text{Actual Air} = \text{Total Air Space} - \text{Machinery (300m}^3\text{)} - \text{Workers (20} \times \text{0.08m}^3\text{)}$$



Surface Area vs. Volume: The Showdown

Feature	Surface Area 	Volume 
What is it?	Paintable outside 	3D Space inside
Joined Shapes	Add CSAs (Faces disappear!) 	Add Total Volumes
Scooped Shapes	ADD the scooped CSA! (More area to paint)	SUBTRACT the scooped Volume! (Less space)
Hidden Faces	They vanish! 	They don't matter! 







Don't forget your basics! 📌

Save for later!



Cube

Volume: a^3
TSA: $6a^2$



Cuboid

Volume: lbh
TSA: $2(lb+bh+hl)$



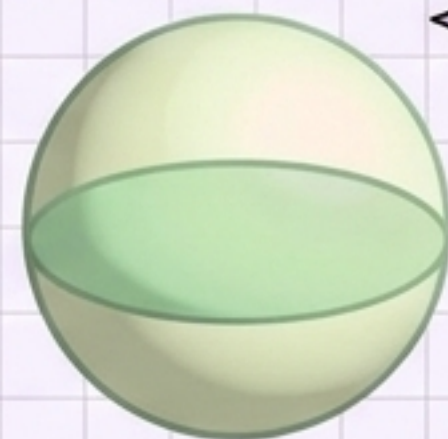
Cylinder

Volume: $\pi r^2 h$
CSA: $2\pi r h$



Cone

Volume: $\frac{1}{3} \pi r^2 h$
CSA: $\pi r l$



Sphere

Volume: $\frac{4}{3} \pi r^3$
TSA: $4\pi r^2$

Save this post to ace your next math exam! 💖🎓